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Editorial Note: This manuscript has been previously reviewed at another journal that is not operating a transparent peer review scheme. This document only contains reviewer comments and rebuttal letters for versions considered at *Nature Communications*.

REVIEWER COMMENTS

Reviewer #2 (Remarks to the Author):

The authors present a comprehensive radiogenomic analysis of a large group pLGG attempting to find immunological characteristics of a pLGG group with worse prognosis, which can already be detected radiologically. It is a very relevant research topic since this pLGG group with worse prognosis is still difficult to define and needs better treatment options. Looking into underlying enriched pathways using RNAsequencing data to identify the patients with disease progression, and correlating that to radiological data for clinical applicability is a novel and valuable approach in the field. Compared to the first submission, the authors made important changes that improved the manuscript. They were able to amend or explain the major concerns. The workflow of different experiments and different cohorts is now more clear.

There are still some questions and minor adjustments needed.

1. Abstract: I think the following first part of this sentence is correct and provides a very strong message about the findings in the paper: "This is the first radiogenomic analysis in pLGGs that enhances prognostication by prediction of immunological profiles, assessment of patients' risk of progression, prediction of treatment response to standard-of-care therapies....."

but I think the last part is too far fetched ", and early stratification of patients to identify potential candidates for novel therapies targeting specific pathways."

I would delete the last part and rephrase it into something like: "These findings could aid in early stratification of high risk pLGG patients, which might benefit from novel therapies targeting pathways that are enriched in the high risk immunogenic group."

2. Introduction line 96-99: "By integrating imaging-derived phenotypes of pLGG with genotypic traits discerned from transcriptional analysis, our goal is to offer an in-depth characterization of pLGGs, "aiding the implementation of precision therapies."

Here the same comment as under 1. I don't think you provide the data for such claim (aiding implementation of precision therapies). Your data can contribute to risk stratification, and based on the transcriptomic data of the high risk immunogenic group potentially novel (precision) therapies could be developed, but you are nowhere near implementing precision therapies since these therapies are not even developed yet.

3. In figure 1 they show the stepwise approach of 1) immunological cluster analysis on n=494, 2) radioimmunomic signature on n=150 and 3) clinicroadiomic risk stratification on n=201 using the radioimmunomic signature developed under 2.

a. The term immune cell deconvolution is not the proper term. It was adjusted in the text but should also be adjusted in the figure.

b. Apologies if I missed it, but I was wondering if the risk prediction model in figure 1D could be further improved when using clinical data + radiomic features + immunological clustering data in the subset of 150 patients?

4. Line 112-114 (We analyzed retrospective data from 545 patients (See Table 1 for detailed description) in the Children's Brain Tumor Network (CBTN 13), encompassing 494 with multi-omic (RNA-seq and

WGS) data...”) and also referring to my previous comment 3b:

what I mean is: if there is RNAseq and WGS data of 490 patients that are described in table 1, then I would expect that a molecular driver is available for all or most of these 490 patients. In principle, all LGG are driven by a molecular alteration that should be detected by WTS and/or WGS. If not then the bioinformatic pipeline clearly is in default. For example: n=16 SEGA wildtype is unusual, since you expect TSC mutations. So probably the TSC gene was not properly covered or filtered out. The same holds true for the remainder ‘wildtype’ cases. (n=80 LGG wildtype + n=24 GNG wildtype + n=16 SEGA wildtype + n=51 not available = 171 of 490 = 35% of the cohort). If you have such extensive molecular tests available, that are able to cover the whole genome/transcriptome and thereby could cover all molecular drivers, I think that 35% of missing data is a lot. Furthermore, it could mean that a (substantial?) part of these cases are misclassified as LGG (since you do not have the molecular driver that supports the LGG diagnosis) and should not be part of this cohort.

I understand if this is not something that you can easily solve, which also doesn’t seem necessary for the way the experiments are designed and conducted. But a small part in the discussion could be added about the limitations that are the consequence of these missing data, as authors have already provided, is indeed the proper thing to do. Also, it is good to be aware of possible misjudgments: First of all, based on the paper by Ryall and al 2020 (PMID: 32289278), it is stated that the type of molecular driver divides the pLGG cohort in 2 different risk groups. It would be very interesting to know whether your transcriptome data could strengthen that or, on the contrary, overrule that. The authors do study this in figure 3F, but since 35% of the cases could not be properly classified, it is the question how that would influence these data.

Secondly, knowing the molecular driver is an important part of the integrated histomolecular diagnosis. For clinicians without access to transcriptomic data or radiomic tools it would also be interesting to know whether a specific type of LGG and / or specific type of molecular driver is also related to the immunogenic high risk group.

In figure 3B you nicely analyse the effect of different LGG molecular subtypes as well as the immuno cluster ID on PFS, but maybe results would have been different if the data of molecular drivers in the cohort would be more complete. For example, now the analysis of LGG wildtype is useless, since it may in fact encompass cases that could be attributed to other LGG classes, of which a statistically significant effect on PFS could be reached when high enough numbers are met.

a. LGG wildtype is not proper nomenclature according to the WHO. In principle, all LGG are driven by a molecular alteration of the MAPK pathway. So if no molecular alteration is known it just means it is not detected. Then the LGG should be termed NOS (if no sequencing tests were performed) or NEC (if sequencing tests did not reveal any results).

b. LGG in general is not proper nomenclature according to the WHO. The diagnosis should be pilocytic astrocytoma, ganglioglioma, RGNT, DNET and so forth. If this information is not available, please mark that as limitation in the paper.

5. Regarding lines 153-173/Figure 3F: cluster 2 encompasses many wildtype cases ◊ might these cases be in fact not LGG but HGG or other types of tumors which explains the lower PFS?

6. Line 149-151: what are the actual TMB numbers per cluster?

7. Line 153-156 / Figure 3F: “related to the WHO brain tumor entities” ◊ there is not 1 proper WHO

brain entity described in this figure 3F (nor in fig 2A, 3B etc).

8. Lastly, the authors show a vast amount of work with impressive plots in fig 3 & 5, but do the authors also provide any pragmatic tools by which these promising prediction tools can be easily implemented in daily clinical practice?

Reviewer #4 (Remarks to the Author): Expert in machine learning and radiology

This study serves as a large radiomic analysis of pediatric low grade glioma. pLGG is relatively rare disease with significant heterogeneity and arguably a good disease to develop improved prognostic models. Ultimately I agree with the authors that multimodal modeling is likely a good method to approach this topic. Overall the strengths of this study are 1) the multimodal nature of the study which spans both imaging and bio- informatics disciplines. 2) the number of experiments which the authors have conducted which do appear to try to unpack the relationship between imaging/immune response in this patient population. The largest overlying weakness in my opinion in the sample size of patients within the imaging cohort. Specifically the discovery cohort is very small and I am not sure that we can reliably draw conclusions from such a small number of patients. Additionally, the relatively incompleteness within the imaging data, I have some concerns regarding how this was methodologically handled. In line with these weaknesses, I have the following more detailed comments.

1) Given the number of sources of data it is difficult to understand how many patients were actually evaluated with a full set of images. I believe it was 201 patients which would make the sample size relatively small (and a validation cohort of 41 is relatively small as well). I would make this a bit more clear in the study design and recognize that limitation within the manuscript.

2) 28 of the patients have evidence of multifocal disease, there is no discussion of how that is being evaluated/handle with respect to radiomic feature analysis. There are a number of publications which have attempted to address this. <https://doi.org/10.1038/s41598-021-89114-6>. Regardless given this is a fair number of patients in their training/validation cohort, it should be described more clearly.

3) It would be valuable to provide imaging information MRI strength, manufacturers between training and testing cohorts. More detail regarding imaging protocols if available should be provided. If not available, this should be discussed as a limitation.

4) The data seems very incomplete with respect to the images and I'm not sure how they handled this in their radiomic analysis. From my view it appears that only 150 patients actually have complete data with all imaging sequences (n=150). How did they handle the additional samples without sequences. These appear to be used in their prediction of tumor progression (n=201)?

5) I am not completely clear why the authors did 2 forms of dimensionality reduction for the radiomic

feature analysis. It appears that they reduced the features using pairwise pearson correlation and then using MRMR feature selection. This could potentially cause issues/bias with the radiomic imaging signal.

6) Figure 5 should demonstrate at risk number for both training and validation cohorts within Kaplan meier curves. Additionally, there should be demonstration of how many patients are in each risk group within panel C/D.

7) In line with previous comments, I feel it would be valuable to provide a detailed table 1 for each of the subgroups described in Figure 1 (immune clusters, imaging group, and risk stratification group).

Reviewer #4 (Remarks on code availability):

Appears to be reasonable.

REVIEWER COMMENTS

Reviewer #2 (Remarks to the Author):

The authors present a comprehensive radiogenomic analysis of a large group pLGG attempting to find immunological characteristics of a pLGG group with worse prognosis, which can already be detected radiologically. It is a very relevant research topic since this pLGG group with worse prognosis is still difficult to define and needs better treatment options. Looking into underlying enriched pathways using RNAsequencing data to identify the patients with disease progression, and correlating that to radiological data for clinical applicability is a novel and valuable approach in the field. Compared to the first submission, the authors made important changes that improved the manuscript. They were able to amend or explain the major concerns. The workflow of different experiments and different cohorts is now more clear. There are still some questions and minor adjustments needed.

Response: We appreciate the helpful feedback provided by this reviewer. We have addressed the concerns and comments, as detailed below.

1. Abstract: I think the following first part of this sentence is correct and provides a very strong message about the findings in the paper: “This is the first radiogenomic analysis in pLGGs that enhances prognostication by prediction of immunological profiles, assessment of patients’ risk of progression, prediction of treatment response to standard-of-care therapies.....”

but I think the last part is too far fetched “, and early stratification of patients to identify potential candidates for novel therapies targeting specific pathways.”

I would delete the last part and rephrase it into something like: “These findings could aid in early stratification of high risk pLGG patients, which might benefit from novel therapies targeting pathways that are enriched in the high risk immunogenic group.”

Response: Thank you, we have revised the abstract:

“These findings could aid in early stratification of high-risk pLGG patients, who may benefit from novel therapies targeting pathways enriched in the high-risk immunological group.”

2. Introduction line 96-99: “By integrating imaging-derived phenotypes of pLGG with genotypic traits discerned from transcriptional analysis, our goal is to offer an in-depth characterization of pLGGs, “ aiding the implementation of precision therapies.”

Here the same comment as under 1. I don’t think you provide the data for such claim (aiding implementation of precision therapies). Your data can contribute to risk stratification, and based on the transcriptomic data of the high risk immunogenic group potentially novel (precision) therapies could be developed, but you are nowhere near implementing precision therapies since these therapies are not even developed yet.

Response: Thank you, we have revised the introduction accordingly:

“By integrating imaging-derived phenotypes of pLGG with genotypic traits discerned from transcriptional analysis, our goal is to offer an in-depth characterization of pLGGs, contributing to risk stratification and identification of high-risk immunogenic patients who may benefit from novel precision therapies.”

3. In figure 1 they show the stepwise approach of 1) immunological cluster analysis on n=494, 2) radioimmunomic signature on n=150 and 3) clinicoradiomic risk stratification on n=201 using the radioimmunomic signature developed under 2.

a. The term immune cell deconvolution is not the proper term. It was adjusted in the text but should also be adjusted in the figure.

Response: Thank you for your note, we revised Figure 1 and replaced the term “deconvolution” with “enrichment”.

b. Apologies if I missed it, but I was wondering if the risk prediction model in figure 1D could be further improved when using clinical data + radiomic features + immunological clustering data in the subset of 150 patients?

Response: Thank you for your comment. We initially did not include the analysis involving immune clusters because our primary objective was to develop a non-invasive predictive marker through clinicoradiomic analysis (using treatment-naïve standard MRI sequences and clinical and demographic variables). This marker aims to inform treatment responsiveness and the risk of progression for patients upfront, allowing us to predict the risk under various scenarios of surgical intervention (e.g., no surgery, biopsy, partial resection, gross/near total resection) and treatment modalities (e.g., no chemotherapy or radiation therapy, chemotherapy, radiotherapy).

Nevertheless, to address the reviewer's comment, we have added this analysis to our Supplementary Information Section 6.3.4. This cohort included 150 patients, with 120 in the discovery set and 30 in the replication set (see Table 1 in the main manuscript). The model (Model 4: Clinical Variables + Radiomics + Immunological Clusters) demonstrated comparable performance to the clinicoradiomic model (Model 2: Clinical Variables + Radiomics), achieving HAR/UNO/IBS indices of 0.76/0.78/0.14 compared to 0.77/0.80/0.16 for the clinicoradiomic model.

Given that immune cluster 2 was associated with a poorer prognosis compared to the clusters 1 and 3, we developed another model with immune cluster 2 and a combination of immune clusters 1 and 3. As shown in the Supplemental Table 8, the performance of this model improved compared to our clinicoradiomic model, with HAR/UNO/IBS indices reaching 0.80/0.83/0.13 in the replication set.

We have added these results to the main manuscript section 2.4:

“Finally, considering the differences in progression-free survival probability between immunological clusters, we evaluated the performance of a model incorporating immunological clusters as a categorical variable (with three distinct values), along with clinical and radiomic variables. This model showed comparable performance to the clinicoradiomic model, achieving Harrell’s concordance indexes of 0.71 [95% CI: 0.58, 0.83] in the discovery set and 0.76 in the replication set, Uno’s concordance indexes of 0.70 [95% CI: 0.60, 0.81] in the discovery set and 0.78 in the replication set, and an IBS of 0.23 [95% CI: 0.16, 0.30] in the discovery set and 0.14 in the replication set.

Given that immune cluster 2 was associated with a poorer prognosis compared to clusters 1 and 3, we trained a Cox model incorporating clinical and radiomic variables, along with a binary variable for immune clusters. In this model, immune cluster 2 was assigned one value, while clusters 1 and 3 were combined and assigned another value. As shown in Supplemental Table 8, this model's performance improved in the replication set compared to our clinicoradiomic model, with Harrell’s concordance index of 0.80, Uno’s concordance index

of 0.83, and an IBS of 0.13. These results suggest that incorporating immunological clustering significantly enhances risk stratification in Cox regression analysis, with immune clusters being a significant variable in these predictive models.”

4. Line 112-114 (We analyzed retrospective data from 545 patients (See Table 1 for detailed description) in the Children’s Brain Tumor Network (CBTN 13), encompassing 494 with multi-omic (RNA-seq and WGS) data…) and also referring to my previous comment 3b:

what I mean is: if there is RNAseq and WGS data of 490 patients that are described in table 1, then I would expect that a molecular driver is available for all or most of these 490 patients. In principle, all LGG are driven by a molecular alteration that should be detected by WTS and or WGS. If not then the bioinformatic pipeline clearly is in default. For example: n=16 SEGA wildtype is unusual, since you expect TSC mutations. So probably the TSC gene was not properly covered or filtered out. The same holds true for the remainder ‘wildtype’ cases. (n=80 LGG wildtype + n=24 GNG wildtype + n=16 SEGA wildtype + n=51 not available = 171 of 490 = 35% of the cohort). If you have such extensive molecular tests available, that are able to cover the whole genome/transcriptome and thereby could cover all molecular drivers, I think that 35% of missing data is a lot. Furthermore, it could mean that a (substantial?) part of these cases are misclassified as LGG (since you do not have the molecular driver that supports the LGG diagnosis) and should not be part of this cohort.

I understand if this is not something that you can easily solve, which also doesn’t seem necessary for the way the experiments are designed and conducted. But a small part in the discussion could be added about the limitations that are the consequence of these missing data, as authors have already provided, is indeed the proper thing to do. Also, it is good to be aware of possible misjudgments: First of all, based on the paper by Ryall and al 2020 (PMID: 32289278), it is stated that the type of molecular driver divides the pLGG cohort in 2 different risk groups. It would be very interesting to know whether your transcriptome data could strengthen that or, on the contrary, overrule that. The authors do study this in figure 3F, but since 35% of the cases could not be properly classified, it is the question how that would influence these data.

Secondly, knowing the molecular driver is an important part of the integrated histomolecular diagnosis. For clinicians without access to transcriptomic data or radiomic tools it would also be interesting to know whether a specific type of LGG and / or specific type of molecular driver is also related to the immunogenic high risk group.

In figure 3B you nicely analyse the effect of different LGG molecular subtypes as well as the immuno cluster ID on PFS, but maybe results would have been different if the data of molecular drivers in the cohort would be more complete. For example, now the analysis of LGG wildtype is useless, since it may in fact encompass cases that could be attributed to other LGG classes, of which a statistically significant effect on PFS could be reached when high enough numbers are met.

Response: We understand the reviewer’s concern regarding a perceived lack of molecular drivers for a substantive proportion of pLGGs. However, our molecular classification of pLGGs is based on our previously published work of the Open Pediatric Brain Tumor Atlas (Open PBTA doi: [10.1016/j.xgen.2023.100340](https://doi.org/10.1016/j.xgen.2023.100340)). This work was documented in a public-facing GitHub repository (<https://github.com/AlexsLemonade/OpenPBTA-analysis>) containing our algorithm for subtyping pLGGs. The molecular classifications for pLGGs were inspired by Ryall, et al (doi: [10.1016/j.ccell.2020.03.011](https://doi.org/10.1016/j.ccell.2020.03.011)) that established prognostic subgroups of pLGGs based on genomic and transcriptomic data. Figure 2A of the work by Ryall, et al highlights the molecular subgroups that best informed risk stratification, which formed the basis for the pLGG annotations in our work.

However, to completely address the reviewer’s concern, we leveraged information regarding molecular drivers in combination with the pathological diagnoses that are collected as part of the CBTN, which

confirmed that all tumors were pLGG entities, to derive 2021 WHO molecular classifications. We have now updated Table 1, all heatmaps, survival models, and association tests in Figures 2, 3, Supplementary Figures 3 & 4, and Supplementary Table 3 reflecting the new 2021 WHO mappings that we derived with associated interpretations in the main text. Notably, a subset (n=47) could only be clearly classified as pediatric-type diffuse low-grade gliomas, NOS based on the nature of the pathology diagnoses that we received from external institutions. Our conclusions regarding the effects of immune cluster and tumor inflammation signature do not change with the update.

a. LGG wildtype is not proper nomenclature according to the WHO. In principle, all LGG are driven by a molecular alteration of the MAPK pathway. So, if no molecular alteration is known it just means it is not detected. Then the LGG should be termed NOS (if no sequencing tests were performed) or NEC (if sequencing tests did not reveal any results).

Response: We understand the reviewer's concern and per our above comment, we have now updated Table 1, all heatmaps, survival models, and association tests in Figures 2, 3, Supplementary Figures 3 & 4, and Supplementary Table 3 reflecting the new 2021 WHO mappings that we derived with associated interpretations in the main text

b. LGG in general is not proper nomenclature according to the WHO. The diagnosis should be pilocytic astrocytoma, ganglioglioma, RGNT, DNET and so fort. If this information is not available, please mark that as limitation in the paper.

Response: Please refer to our responses in (4) and (4a). We have now addressed this limitation by mapping to official WHO 2021 tumor classifications.

5. Regarding lines 153-173/Figure 3F: cluster 2 encompasses many wildtype cases, might these cases be in fact not LGG but HGG or other types of tumors which explains the lower PFS?

Response: We have leveraged pathological diagnoses from CBTN in combination with molecular driver information to map to 2021 WHO pLGG entities. We can confirm that all cases were in fact LGG. The Cox regression model in figure 3B has now been updated with these entities and cluster 2 remains with a significantly higher overall hazard relative to clusters 1 and 3. Further, figure 3F has now been updated to a correspondence analysis to show the association between immune clusters and 2021 WHO entities.

6. Line 149-151: what are the actual TMB numbers per cluster?

Response: For all clusters, the lower quartile, median, and upper quartile for TMB have now been added for all groups.

7. Line 153-156 / Figure 3F: " related to the WHO brain tumor entities" – there is not 1 proper WHO brain entity described in this figure 3F (nor in fig 2A, 3B etc).

Response: We have now addressed this by leveraging molecular and pathological diagnostic information from CBTN to map to proper WHO CNS tumor entities. Figure 3F has now been updated with the proper terminology.

8. Lastly, the authors show a vast amount of work with impressive plots in fig 3 & 5, but do the authors also provide any pragmatic tools by which these promising prediction tools can be easily implemented in daily clinical practice?

Response: Thank you for your comment. We acknowledge the importance of providing pragmatic and user-friendly tools for easy application in clinical practice. Some of the analytical approaches described in this paper are already incorporated and routinely used in the CBTN workflows. Our automated image preprocessing and tumor segmentation (which is made publicly accessible: <https://github.com/d3b-center/peds-brain-seg-pipeline-public>, <https://github.com/d3b-center/peds-brain-auto-seg-public>). These tools along with our radiomic feature extraction pipeline are available as gears incorporated with Flywheel, which is a CBTN imaging repository, and made available to the CBTN investigators to seamlessly preprocess MRI scans, segment tumor subregions, and extract radiomic features. Flywheel gears for predicting the risk scores and groups, as well as immune cluster assignments are underway. Similarly, the bioinformatic analysis pipelines, can be accessed publicly through the CBTN's CAVATICA platform and the processed data is made available through OpenPedCan. We will continue to make these tools more user-friendly for easier clinical incorporation.

Reviewer #4 (Remarks to the Author): Expert in machine learning and radiology

This study serves as a large radiomic analysis of pediatric low grade glioma. pLGG is relatively rare disease with significant heterogeneity and arguably a good disease to develop improved prognostic models. Ultimately I agree with the authors that multimodal modeling is likely a good method to approach this topic. Overall the strengths of this study are 1) the multimodal nature of the study which spans both imaging and bio-informatics disciplines. 2) the number of experiments which the authors have conducted which do appear to try to unpack the relationship between imaging/immune response in this patient population. The largest overlying weakness in my opinion in the sample size of patients within the imaging cohort. Specifically the discovery cohort is very small and I am not sure that we can reliably draw conclusions from such a small number of patients. Additionally, the relatively incompleteness within the imaging data, I have some concerns regarding how this was methodologically handled. In line with these weaknesses, I have the following more detailed comments.

Response: We appreciate the helpful feedback provided by this reviewer. We acknowledge that our imaging cohort ($n = 201$ for clinoradiomic analysis and $n = 150$ for radioimmunomic analysis) is relatively small. However, it is important to note that acquiring a significantly larger cohort with the necessary imaging, genomic, and RNA-seq data is currently challenging. This is particularly true for pediatric low-grade glioma, which is rare compared to adult gliomas, and even for adult glioblastoma, one of the most studied brain tumors.

Recognizing this concern, at the Children's Brain Tumor Network (CBTN), we are actively working to increase our multi-modal sample size by collaborating with other institutions both nationally and internationally. This is however, an extensive and lengthy process.

In terms of methodology, we have taken rigorous precautions in training our machine learning models. We split the available data into a "discovery" set for training and validation using a thorough nested cross-

validation approach. Additionally, we used a “replication” set, which remained fully independent and unseen during model training, to assess the generalizability of our trained models to new data.

1) Given the number of sources of data it is difficult to understand how many patients were actually evaluated with a full set of images. I believe it was 201 patients which would make the sample size relatively small (and a validation cohort of 41 is relatively small as well). I would make this a bit more clear in the study design and recognize that limitation within the manuscript.

Response: Thank you. To address your comments, we have now revised Table 1 to better reflect the breakdown of characteristics for the patients in each of the imaging analyses cohorts. Furthermore, Figure 1 shows the number of patients with imaging data used in each set of analysis; for more clarity, in section D, we added the number of subjects used for clinoradiomic analysis (n=201) and transcriptomic analysis for identification of Biological and Immune Pathways Correlated with Clinoradiomic Risk (n = 150).

We have also revised the study design for more clarity: “The imaging cohort for clinoradiomic risk stratification analysis, consisting of 201 patients, was randomly divided into two groups: a 'discovery' cohort with 160 patients for training and validation of the predictive models, and an 'unseen replication' cohort with 41 patients for independent testing of the trained models. For the radioimmunomic analysis, a supervised classification approach was used, which required the availability of both imaging and transcriptomic data. Maintaining the same 'discovery' and 'replication' cohort assignments, 150 subjects were found to have both imaging and transcriptomic data. Of these, 120 patients were included in the 'discovery' set, and 30 patients were included in the 'replication' set.”

We had initially mentioned this limitation: “The current study encountered challenges due to the limited availability of imaging and genomic data (WGS or RNA-sequencing), compounded by the rarity of pediatric brain tumors.”

To address this reviewer’s comment, we also added the following:

“We took rigorous precautions in training our machine learning models by splitting the available data into a “discovery” set for training and validation, using a thorough nested cross-validation approach. Additionally, we employed a fully independent “replication” set, unseen during model training, to assess the generalizability of our models to new data. Despite these measures, our imaging cohort was relatively small and included data from only a single institution, highlighting the need for further external validation to test its generalizability”.

2) 28 of the patients have evidence of multifocal disease, there is no discussion of how that is being evaluated/handle with respect to radiomic feature analysis. There are a number of publications which have attempted to address this. <https://doi.org/10.1038/s41598-021-89114-6>. Regardless given this is a fair number of patients in their training/validation cohort, it should be described more clearly.

Response: Thank you for bringing this up. To clarify our imaging cohort, as indicated in our previous Supplementary Table 1 and the revised Table 1 (see Table 1 in our revised manuscript), the number of patients classified as multifocal was 14. As described in our Methods section 4.1.1, our exclusion criteria specifically encompassed the situations where the primary site was not the brain or leptomeningeal dissemination was reported, ensuring that none of the tumors were metastatic.

Upon reviewing the MRI data for the subjects categorized as multifocal in our imaging cohort again, we confirmed that these tumors do not include multiple distinct foci, but rather encompass multiple regions of the brain. For example, a tumor in the frontal region extending into the ventricles, a tumor covering the temporal lobe and basal ganglia, a tumor in the optic pathway extending into the thalamus or temporal lobe, or a tumor covering multiple lobes of the brain. These are contiguous tumors extending into multiple regions.

To facilitate Cox regression in our clinicoradiomic analysis, we consolidated tumors covering more than one of the other main categories into one group. This avoids complications in the analysis due to having too many categories with a small number of subjects per category. To avoid any confusions, we changed the term from “multifocal” to “multi-regional” in Table 1.

The following table shows the locations of these tumors, extracted from the metadata provided through the CBTN database:

SubjectID	Tumor Locations	Metastasis
C1077357	Brainstem- Midbrain/Tectum; Thalamus	No
C66420	Optic Pathway; Temporal Lobe	No
C23616	Thalamus; Ventricles	No
C1072314	Basal Ganglia; Temporal Lobe; Thalamus	No
C2734536	Basal Ganglia; Temporal Lobe	No
C719919	Brainstem- Midbrain/Tectum; Thalamus	No
C60270	Brainstem- Midbrain/Tectum; Thalamus	No
C50676	Brainstem-Pons; Cerebellum/Posterior Fossa	No
C1046730	Brainstem-Pons; Cerebellum/Posterior Fossa	No
C3365895	Brainstem-Pons; Brainstem-Midbrain/Tectum	No
C24969	Brainstem-Medulla; Brainstem-Midbrain/Tectum; Brainstem-Pons	No
C53505	Brainstem-Medulla; Brainstem-Midbrain/Tectum; Cerebellum/Posterior Fossa	No
C324351	Optic Pathway; Temporal Lobe	No
C377733	Optic Pathway; Basal Ganglia; Brainstem-Midbrain/Tectum	No

The paper cited by the reviewer focuses on brain metastases, where multiple disjoint lesions exist and can be considered as separate and distinct for radiomic analysis. However, as explained earlier, we included primary tumors without any metastasis or multiple disjoint lesions.

3) It would be valuable to provide imaging information MRI strength, manufacturers between training and testing cohorts. More detail regarding imaging protocols if available should be provided. If not available, this should be discussed as a limitation.

Response: Thanks for your comment. We have included the magnetic field strength and manufacturers between the discovery and replication sets in the Supplementary Table 2 and 10.

4) The data seems very incomplete with respect to the images and I'm not sure how they handled this in their radiomic analysis. From my view it appears that only 150 patients actually have complete data with all imaging sequences (n=150). How did they handle the additional samples without sequences. These appear to be used in their prediction of tumor progression (n=201)?

Response: As mentioned earlier, we have taken rigorous precautions in training our machine learning models. We split the available data into a “discovery” set for training and validation using a thorough nested cross-validation approach. Additionally, we used a “replication” set, which remained fully independent and unseen during model training, to assess the generalizability of our models to new data.

The radioimmunomic signature, which predicts immunological clusters using radiomic features, was generated from a cohort of 150 subjects who had both MRI and RNA-seq data.

For risk stratification (prediction of tumor progression), our clinicroadiomic model only includes clinical and MRI data, which were available for all 201 subjects. When exploring the associations of risk groups with immunological clusters and transcriptomic pathways in section 2.5 (see updated Figure 1D), we used data from the intersection of the clinicroadiomic and transcriptome cohorts (150 patients).

We provided a subcohort analysis for these 150 patients in the Supplementary Information to demonstrate the added value of molecular subtypes or immunological clusters to the clinicroadiomic features. These subcohort analyses were performed only on subjects with available clinical, MRI, and molecular data.

5) I am not completely clear why the authors did 2 forms of dimensionality reduction for the radiomic feature analysis. It appears that they reduced the features using pairwise pearson correlation and then using MRMR feature selection. This could potentially cause issues/bias with the radiomic imaging signal.

Response: In our study, an approach using both pairwise Pearson correlation (feature filtering) and MRMR feature selection (wrapper) for the radiomic feature analysis was adopted due to their complementary strengths. These two methods mitigate different types of redundancy. Pairwise Pearson correlation analysis between radiomic features in a filtering approach (without the use of their relationship with the target outcome) was used to identify and remove features that are highly correlated with each other, reducing redundancy. This step helps prevent multicollinearity issues, ensuring that the remaining features are not linearly dependent.

MRMR feature selection selects features that are both relevant to the target outcome and minimally redundant. It goes beyond simple correlation of radiomic features with each other, by considering the mutual information between features and the target variable, capturing both linear and non-linear relationships. By initially applying Pearson correlation, we simplify the dataset by removing highly correlated features, making the subsequent MRMR analysis more efficient and focused. MRMR then further refines the feature set by selecting the most relevant features with minimal redundancy, ensuring that the final set of features is both diverse and highly informative.

Validation of our model trained using nested cross-validation and independently tested on an “unseen” set, demonstrated that the combined approach is stable and unbiased.

6) Figure 5 should demonstrate at risk number for both training and validation cohorts within Kaplan Meier curves. Additionally, there should be demonstration of how many patients are in each risk group within panel C/D.

Response: Thank you for your note. We modified the Kaplan-Meier curves in Figure 5A, and the Association and Sankey diagrams in Figure 5C-D.

7) In line with previous comments, I feel it would be valuable to provide a detailed table 1 for each of the subgroups described in Figure 1 (immune clusters, imaging group, and risk stratification group).

Response: Thank you, we have updated Table 1 to include the specific details of the cohorts included in the analyses of immunological clusters, radioimmunomic signature, and clinicoradiomic risk stratification.

Reviewer #4 (Remarks on code availability):

Appears to be reasonable.

REVIEWERS' COMMENTS

Reviewer #2 (Remarks to the Author):

All my concerns have been addressed in the revisions by the authors. I hereby support publication of this work in Nature Communications.

Reviewer #4 (Remarks to the Author):

Overall I feel the authors have attempted to address a majority of the concerns in the review. I do feel that the authors have largely addressed the comments by reviewers. I do feel the major critique is still related to cohort derivation and sample size.

1) The cohort derivation appears to be very particular with specific exclusions in what appears to be reasonable situations. I do have concerns that this may affect the overall generalizability of their findings.

2) Re: comment 6 from reviewer 4. The at-risk numbers in figure 3a do show that a majority of the cohort die in the first 2000 days. I recommend focusing the KM curve on 5000 days and providing a more granular view of the number at risk (every 500 days). This provides more clarity.

3) Re: Comment 6 from reviewer 4: The evaluated risk for the clinical radiomic KM analysis (figure 5a-b) are extremely small and certainly not reasonable to show a true benefit to the model. The validation cohort is a total of 41 patients across 3 groups which is very limited.

Reviewer #4 (Remarks on code availability):

only reviewed the readme- but seems reasonable overall.

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1) The cohort derivation appears to be very particular with specific exclusions in what appears to be reasonable situations. I do have concerns that this may affect the overall generalizability of their findings.

Response: We acknowledge this limitation in our discussion: "The current study encountered challenges due to the limited availability of imaging and genomic data (WGS or RNA-sequencing), compounded by the rarity of pediatric brain tumors. We took rigorous precautions in training our machine learning models by splitting the available data into a "discovery" set for training and validation, using a thorough nested cross-validation approach. Additionally, we employed a fully independent "replication" set, unseen during model training, to assess the generalizability of our models to new data. Despite these measures, our imaging cohort was relatively small and included data from only a single institution, highlighting the need for further external validation to test its generalizability."

2) Re: comment 6 from reviewer 4. The at-risk numbers in figure 3a do show that a majority of the cohort die in the first 2000 days. I recommend focusing the KM curve on 5000 days and providing a more granular view of the number at risk (every 500 days). This provides more clarity.

Response: Thanks for your comment, in our revised version, we have updated the KM curve in Figure 3A to address this comment.

3) Re: Comment 6 from reviewer 4: The evaluated risk for the clinical radiomic KM analysis (figure 5a-b) are extremely small and certainly not reasonable to show a true benefit to the model. The validation cohort is a total of 41 patients across 3 groups which is very limited.

Response: We acknowledge this limitation in our discussion; please see our response to your comment #1.

Reviewer #4 (Remarks on code availability):

only reviewed the readme- but seems reasonable overall.